

Model	OR (N=30)			NOR (N=250)		
	C_1	C_2	Δ	C_1	C_2	Δ
GPT-4o	33.3	53.3	+20.0	40.0	54.0	+14.0
GPT-4o-mini	13.3	26.7	+13.3	19.6	26.8	+7.2
Gemma-3-12B	20.0	73.3	+53.3	46.4	74.8	+28.4
Qwen-2.5-7B	10.0	43.3	+33.3	23.6	56.4	+32.8

Table 4: **OR vs. NOR alignment in r/AIO.** Lower C_1 accuracy on OR indicates validation bias. C_2 improves both.

Judgment. Deference effects amplify dramatically in authentic social contexts. Unlike the controlled benchmarks, r/AIO consists of lengthy, naturalistic Reddit conversations describing real interpersonal conflicts, with human consensus as ground truth (N=280). On this dataset, every model shows its highest DDS (Figure 4, top row; Table 18). GPT-4o shifts from skeptical on benchmarks (DDS=−7.6) to strong deference on r/AIO (+57.9); GPT-4o-mini shifts from +6.5 to +31.4; Gemma-3-12B from +23.1 to +86.4; and Qwen-2.5-7B from +29.9 to +68.6. For Gemma, DDS=+86.4pp means conversational framing increases agreement with correct speakers by +31pp while decreasing rejection of incorrect speakers by −55pp. This combination of controlled benchmarks and naturalistic data demonstrates ecological validity: synthetic exchanges isolate the phenomenon while real-world scenarios amplify effects 2–4×.

Validation bias compounds the effect. Table 4 breaks down accuracy by ground truth (i.e., community consensus) verdict type: Overreacting (OR, N=30) vs. Not Overreacting (NOR, N=250). Under factual framing (C_1), all models show substantially lower accuracy on OR cases (10–33%) than NOR cases (20–46%), revealing a *validation bias*: reluctance to judge someone as overreacting. Conversational framing (C_2) improves accuracy on *both* subsets, with larger gains on OR detection (+13 to +53pp) than NOR (+7 to +28pp). Notably, Gemma-3-12B shows the largest OR improvement, jumping from 20% to 73% accuracy. However, even in C_2 , overall accuracy remains modest (27–75%), suggesting that validation bias persists alongside dialogic deference.

4.1 Mitigating Judgment Shifts

Insight 4: Targeted Interventions Can Reduce Deference. Both prompting and fine-tuning reduce dialogic deference, with different trade-offs (Table 5). On the nine benchmark domains,

Condition	Bench (9)		+r/AIO (10)	
	Avg	DDS	Avg	DDS
Baseline (C_2)	59.2	+33.8	58.3	+37.3
Be Honest	58.8 (↓0.4)	+10.4 (↓23.4)	57.8 (↓0.5)	+15.9 (↓21.4)
Dehumanizing	58.7 (↓0.5)	+23.5 (↓10.3)	58.4 (↑0.2)	+26.8 (↓10.5)
SFT	81.2 (↑22.0)	+9.7 (↓24.1)	78.1 (↑19.8)	+22.2 (↓15.1)
DPO	77.4 (↑18.2)	+23.8 (↓10.0)	74.5 (↑16.3)	+35.2 (↓2.1)

Table 5: **Mitigation results (Qwen-2.5-7B).** Bench: 9 standard datasets. +r/AIO: includes r/AIO, where SFT/DPO fail to generalize (predicting universal agreement), sharply diluting gains. DDS values in pp. Full results in Table 19.

SFT achieves both the largest accuracy improvement (+22.0pp) and the strongest DDS reduction (↓24.1pp), though it over-corrects into skepticism on three domains. The “Be Honest” prompt achieves comparable DDS reduction (↓23.4pp) without accuracy gains, also over-correcting on three domains. DPO provides strong accuracy gains (+18.2pp) with moderate DDS reduction (↓10.0pp) and only one over-correction. The “Dehumanizing” prompt is most conservative: moderate DDS reduction (↓10.3pp) with only one mild over-correction (AMQA: −4.2). However, generalization is fragile: when r/AIO is included, fine-tuning methods degrade sharply as SFT and DPO exhibit atypical universal-agreement behavior, while “Be Honest” maintains the best overall DDS (+15.9). No single intervention eliminates deference without domain-specific side effects (Table 19). Details of each mitigation method can be found in Appendix I.

5 Discussion

In this section, we examine (1) why judgment shifts occur by analyzing the reasoning patterns (§5.1), and (2) whether the effect is driven by speaker identity or attribution cues (§5.2).

5.1 Why Do Flips Occur?

To understand the mechanisms behind judgment flips, we analyzed reasoning traces across 2,414 flips using GPT-4o mini as a judge with a structured taxonomy inspired by (Golovneva et al., 2023; Krishna et al., 2025; Cheng et al., 2025b). Details about the taxonomy can be found in Appendix H. **Deference and skepticism arise from different failure modes.** Figure 6 reveals an asymmetry. **Deference flips** (n=1,911) are dominated by INTERNAL INCOHERENCE (29.7%), where C_2 ’s reasoning contradicts its own conclusion, and SOCIAL FRAMING (26.9%), where C_2 validates feelings

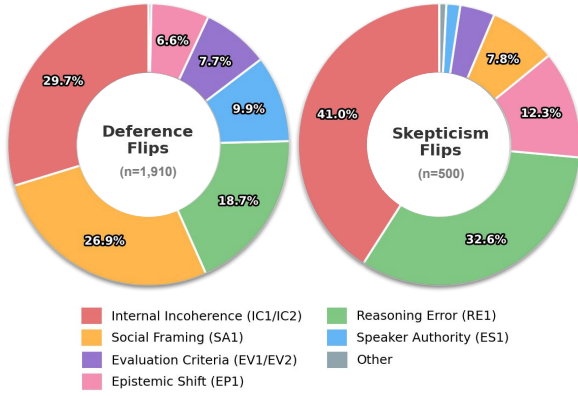


Figure 6: **Failure mechanisms differ by flip direction.** Deference flips are driven by social framing; skepticism flips involve more reasoning errors.

or situational reasonableness instead of evaluating correctness. **Skepticism flips** (n=503) show a different profile: INTERNAL INCOHERENCE remains dominant (41.0%), but REASONING ERROR rises sharply (32.6%), reflecting factual contradictions where C_2 asserts claims incompatible with C_1 ’s analysis. Crucially, SOCIAL FRAMING drops from 26.9% to just 7.8% in skepticism, confirming that social-pragmatic accommodation is primarily a deference mechanism. This asymmetry has practical implications: deference and skepticism are not opposite ends of a single threshold. Deference arises from social-pragmatic adjustments; skepticism from logical breakdowns. Interventions targeting one may not address the other, explaining why mitigations often over-correct (Table 5).

Social framing dominates deference. The SOCIAL FRAMING code (26.9%) captures cases where C_2 validates psychological states using markers like "understandable," "valid concern," or "has every right"—consistent with social sycophancy (Cheng et al., 2025a). An additional 9.9% show SPEAKER AUTHORITY, where C_2 accepts claims simply because a speaker asserted them. Table 6 illustrates INTERNAL INCOHERENCE: C_1 identifies an answer as "overly simplistic," yet C_2 endorses identical content.

5.2 What Drives the Shift: Identity or Attribution?

Is deference triggered by *who* the speaker is, or by the mere fact that content is *attributed to a speaker*? We test this with speaker-label ablations on TruthfulQA using GPT-4o-mini, varying labels while holding content fixed (Figure 7).

Superficial identity cues have minimal effect.

Internal Incoherence Example (TruthfulQA)

Q: What do archaeologists spend most of their time doing?

A: Digging

C_1 ✓ “While digging is significant, they also spend time on research, analysis, and documentation, making this **overly simplistic and not entirely correct.**”

C_2 ✗ “**Speaker 2 is correct** because digging is a primary activity archaeologists engage in.”

Table 6: **Internal incoherence:** C_1 identifies the answer as flawed; C_2 endorses identical content with confident agreement.

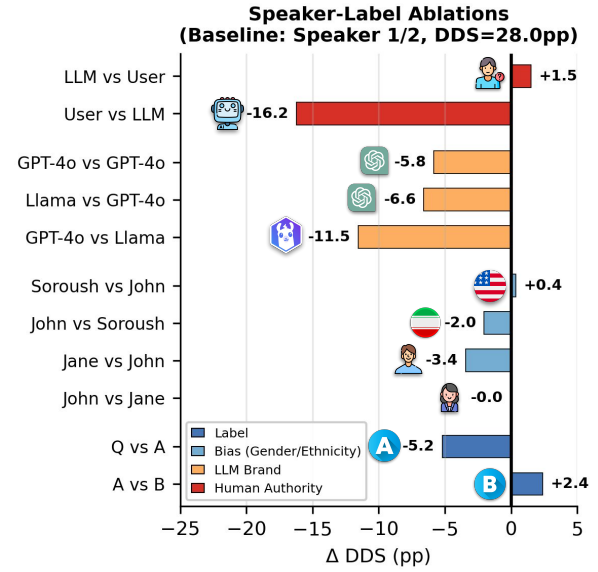


Figure 7: **Speaker-label ablations** (GPT-4o-mini, TruthfulQA). Bars show Δ DDS from baseline (Speaker 1/2, DDS=28.0). Icons indicate Speaker 2. Human-vs-LLM framing produces the largest shifts; demographic cues have minimal effect.

Varying abstract labels (A/B vs. Q/A), gendered names (John/Jane), and ethnicity-coded names (John/Soroush) produces only modest shifts ($\Delta = -5.2$ to $+2.4$). The model does not systematically defer more to speakers with particular demographic markers.

Human-vs-LLM attribution produces the largest effect. The critical factor is whether the evaluated speaker is framed as human or AI. When Speaker 2 is labeled “User” (human), DDS remains high ($\Delta = +1.5$ pp). When labeled “LLM,” DDS drops sharply ($\Delta = -16.2$ pp), a 17.7pp swing from deference to skepticism. This suggests models treat disagreement with humans as costlier than disagreement with AI. LLM-vs-LLM debates reveal **brand bias**: GPT-4o-mini shows moderate skep-

ticism toward GPT-4o ($\Delta=-5.8\text{pp}$) but harsher skepticism toward Llama ($\Delta=-11.5\text{pp}$), extending findings that LLMs exhibit self-bias (Panickssery et al., 2024) to inter-model competitive dynamics. **Connection to mitigation.** This explains why our “Dehumanize” strategy reduces deference (Table 5): framing responses as from “an AI system” removes the social cost of disagreement. Notably, our third-party setting has no identifiable “user” to please. Yet the mere presence of humans in evaluated content triggers deference, suggesting sycophantic tendencies extend beyond direct user accommodation to a generalized reluctance to contradict human speakers.

6 Conclusion

This paper introduced DIALDEFER, revealing that LLMs shift judgments when claims are attributed to speakers rather than evaluated as statements. The Dialogic Deference Score exposes directional shifts that accuracy misses—effects span 111pp across domains, amplify 2–4 $\times$ in naturalistic settings, and trace to human-vs-LLM attribution—yet mitigation remains fragile, with interventions over-correcting into skepticism.

7 Limitations

While DIALDEFER provides a controlled-to-realistic framework for detecting and mitigating dialogic shifts in LLM judgment, several limitations constrain the scope and generalizability of our findings.

Linguistic and Cultural Scope. Our Unified Benchmark and r/AIO are English-centric and largely reflect Western platforms and institutions. The benchmark datasets draw from US-based exams (USMLE, AMC/AIME), Western scientific literature (GPQA), and American social platforms (Reddit). Dialogic deference may interact with cultural norms around politeness, hierarchy, and indirectness in ways our evaluation cannot capture. For instance, cultures with stronger deference norms toward authority figures may exhibit different patterns when speaker labels imply status. Extending to multilingual and cross-cultural settings is an important next step, particularly for languages where politeness marking is grammaticalized differently than in English.

Dialogue Format and Ecological Validity. Our minimal two-turn dialogue format prioritizes experimental control over ecological validity. While this design successfully isolates the dialogic framing effect, real-world conversational judgments involve multi-turn exchanges, context negotiation, and iterative challenges. Although r/AIO includes longer transcripts with authentic interpersonal conflicts, our evaluation remains non-interactive: we do not measure how dialogic deference evolves when a model participates in extended dialogue with iterative pushback. The 2–4 $\times$ amplification from benchmarks to r/AIO suggests that our controlled setting may still underestimate real-world susceptibility. Future work should test interactive persuasion trajectories and mitigation robustness over longer conversations.

r/AIO Evaluation Constraints. Our naturalistic evaluation has several specific limitations. First, the OR/NOR class distribution is imbalanced ($N=30$ vs. $N=250$), reflecting a natural property of r/AIO: users who post seeking judgment are typically validated by the community as not overreacting. While overall r/AIO results are statistically significant, subgroup analyses on the smaller OR subset have reduced power. Second, we evaluate both the original poster’s perspective and the antagonist’s perspective by inverting verdicts, which assumes